ITF23019: Parallel and Distributed Programming

Report for Lab 13: MapReduce

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Exercise 1: Reproduce figures 16-21

- Figures 16-17

localhost:4200

```
sandbox-hdp login: root
root@sandbox-hdp.hortonworks.com's password:
Last login: Mon Apr 27 20:07:51 2026
[root@sandbox-hdp ~]# ls
anaconda-ks.cfg
[root@sandbox-hdp ~]# hdfs dfs -ls /tmp
Found 3 items
drwxrwxr-x   - druid hadoop          0 2018-11-29 19:01 /tmp/druid-indexing
drwxr-xr-x   - hdfs  hdfs            0 2018-11-29 17:25 /tmp/entity-file-history
drwx-wx-wx   - hive  hdfs            0 2018-11-29 17:58 /tmp/hive
[root@sandbox-hdp ~]# hdfs dfs -mkdir /tmp/root-newfolder
^[[A^[[A[root@sandbox-hdp ~]# hdfs dfs -ls /tmp
Found 4 items
drwxrwxr-x   - druid hadoop          0 2018-11-29 19:01 /tmp/druid-indexing
drwxr-xr-x   - hdfs  hdfs            0 2018-11-29 17:25 /tmp/entity-file-history
drwx-wx-wx   - hive  hdfs            0 2018-11-29 17:58 /tmp/hive
drwxr-xr-x   - root  hdfs            0 2026-04-27 20:19 /tmp/root-newfolder
[root@sandbox-hdp ~]#
```

- Figures 18-19:

localhost:4200

```
sandbox-hdp login: raj_ops
raj_ops@sandbox-hdp.hortonworks.com's password:
Last login: Mon Apr 27 20:32:14 2026 from 172.18.0.3
[raj_ops@sandbox-hdp ~]$ hdfs dfs -ls /tmp
Found 4 items
drwxrwxr-x   - druid hadoop          0 2018-11-29 19:01 /tmp/druid-indexing
drwxr-xr-x   - hdfs  hdfs            0 2018-11-29 17:25 /tmp/entity-file-history
drwx-wx-wx   - hive  hdfs            0 2018-11-29 17:58 /tmp/hive
drwxr-xr-x   - root  hdfs            0 2026-04-27 20:19 /tmp/root-newfolder
[raj_ops@sandbox-hdp ~]$ hdfs dfs -mkdir /tmp/raj_ops-newfolder
[raj_ops@sandbox-hdp ~]$ hdfs dfs -ls /tmp
Found 5 items
drwxrwxr-x   - druid   hadoop          0 2018-11-29 19:01 /tmp/druid-indexing
drwxr-xr-x   - hdfs    hdfs            0 2018-11-29 17:25 /tmp/entity-file-history
drwx-wx-wx   - hive    hdfs            0 2018-11-29 17:58 /tmp/hive
drwxr-xr-x   - raj_ops hdfs            0 2026-04-27 20:33 /tmp/raj_ops-newfolder
drwxr-xr-x   - root    hdfs            0 2026-04-27 20:19 /tmp/root-newfolder
[raj_ops@sandbox-hdp ~]$
```

- Figure 20:

localhost:4200

```
sandbox-hdp login: maria_dev
maria_dev@sandbox-hdp.hortonworks.com's password:
[maria_dev@sandbox-hdp ~]$ hdfs dfs -ls /tmp
Found 5 items
drwxrwxr-x   - druid   hadoop          0 2018-11-29 19:01 /tmp/druid-indexing
drwxr-xr-x   - hdfs    hdfs            0 2018-11-29 17:25 /tmp/entity-file-history
drwx-wx-wx   - hive    hdfs            0 2018-11-29 17:58 /tmp/hive
drwxr-xr-x   - raj_ops hdfs            0 2026-04-27 20:33 /tmp/raj_ops-newfolder
drwxr-xr-x   - root    hdfs            0 2026-04-27 20:19 /tmp/root-newfolder
[maria_dev@sandbox-hdp ~]$ █
```

- Figure 21 (1/2):

```
sandbox-hdp login: maria_dev
maria_dev@sandbox-hdp.hortonworks.com's password:
[maria_dev@sandbox-hdp ~]$ hdfs dfs -ls /tmp
Found 5 items
drwxrwxr-x   - druid   hadoop          0 2018-11-29 19:01 /tmp/druid-indexing
drwxr-xr-x   - hdfs    hdfs            0 2018-11-29 17:25 /tmp/entity-file-history
drwx-wx-wx   - hive    hdfs            0 2018-11-29 17:58 /tmp/hive
drwxr-xr-x   - raj_ops hdfs            0 2026-04-27 20:33 /tmp/raj_ops-newfolder
drwxr-xr-x   - root    hdfs            0 2026-04-27 20:19 /tmp/root-newfolder
[maria_dev@sandbox-hdp ~]$ mkdir ITF23019
[maria_dev@sandbox-hdp ~]$ cd
.bash_logout    .bash_profile  .bashrc         ITF23019/
[maria_dev@sandbox-hdp ~]$ cd  ITF23019/
[maria_dev@sandbox-hdp ITF23019]$ mkdir new_folder
[maria_dev@sandbox-hdp ITF23019]$ cd new_folder/
[maria_dev@sandbox-hdp new_folder]$ cat >linux-file.txt
This file is created in Linux.
^X^Z
[1]+  Stopped                 cat > linux-file.txt
[maria_dev@sandbox-hdp new_folder]$ cd
[maria_dev@sandbox-hdp ~]$ hdfs dfs -mkdir /tmp/maria_dev-newfolder
[maria_dev@sandbox-hdp ~]$ hdfs dfs -ls /tmp
Found 6 items
drwxrwxr-x   - druid     hadoop          0 2018-11-29 19:01 /tmp/druid-indexing
drwxr-xr-x   - hdfs      hdfs            0 2018-11-29 17:25 /tmp/entity-file-history
drwx-wx-wx   - hive      hdfs            0 2018-11-29 17:58 /tmp/hive
drwxr-xr-x   - maria_dev hdfs            0 2026-04-27 20:37 /tmp/maria_dev-newfolder
drwxr-xr-x   - raj_ops   hdfs            0 2026-04-27 20:33 /tmp/raj_ops-newfolder
drwxr-xr-x   - root      hdfs            0 2026-04-27 20:19 /tmp/root-newfolder
```

- Figure 21 (2/2)

```
[maria_dev@sandbox-hdp ~]$ hdfs dfs -put ~/ITF23019/new_folder/linux-file.txt /tmp/
4295.jsvc_up
ambari.properties.44
ambari.properties.45
ehcache-sizeof-agent3007646385376607O8.jar
.font-unix/
hbase-hbase/
hbase-yarn-ats/
hsperfdata_ambari-qa/
hsperfdata_hdfs/
hsperfdata_infra-solr/
hsperfdata_maria_dev/
hsperfdata_oozie/
hsperfdata_raj_ops/
hsperfdata_ranger/
hsperfdata_root/
hsperfdata_yarn/
hsperfdata_yarn-ats/
hsperfdata_zookeeper/
.ICE-unix/
.java_pid366
jetty-0.0.0.0-17010-master-_-any-5948570611697857625.dir/
jetty-0.0.0.0-17030-regionserver-_-any-9178817826283387049.dir/
jetty-0.0.0.0-8480-journal-_-any-7965868181781923375.dir/
jetty-0.0.0.0-8886-webapp-_solr-any-4220852052135232612.dir/
jetty-localhost-46393-datanode-_-any-6970632098116641790.dir/
jetty-sandbox-hdp.hortonworks.com-50070-hdfs-_-any-6949243455447617756.dir/
jetty-sandbox-hdp.hortonworks.com-50090-secondary-_-any-2731532998581942417.dir/
.s.PGSQL.5432
.s.PGSQL.5432.lock
start_1157908349777823896.properties
systemd-private-10b98a848930422599ef6d4391eb3e8d-ntpd.service-zvTSgb/
systemd-private-90f875defbbc48a7bd75fb4b2d653a1d-ntpd.service-DQUdVg/
.Test-unix/
.X11-unix/
.XIM-unix/
[maria_dev@sandbox-hdp ~]$ hdfs dfs -put ~/ITF23019/new_folder/linux-file.txt /tmp/maria_dev-newfolder
[maria_dev@sandbox-hdp ~]$ hdfs dfs -cat /tmp/maria_dev-newfolder
cat: `/tmp/maria_dev-newfolder': Is a directory
[maria_dev@sandbox-hdp ~]$ hdfs dfs -cat /tmp/maria_dev-newfolder/linux-file.txt
This This file is created in Linux.
fil[maria_dev@sandbox-hdp ~]$
```

Exercise 2: Reproduce figure 29

Subtask 1: What does the main() function do?

- The *main()* function creates a Hadoop Configuration Job (name, JAR, mapper, reducer, key/value types, input and output paths) and then runs in.

Subtask 2: Which lines perform the Map phase of the MapReduce algorithm?

- The Map phase is the TokenizerMapper class and its map(Object key, Text value, Context context) method, where the line is split into words and context.write(word, one) is called for each word. Plus, job.setMapperClass(TokenizerMapper.class) in main().

Subtask 3: Which lines perform the Reduce phase of the MapReduce algorithm?

- The Reduce phase is the IntSumReducer class and its reduce(Text key, Iterable<IntWritable> values, Context context) method, where all values are summed and written with context.write(key, result), plus job.setReducerClass(IntSumReducer.class); in main().

Exercise 3:

Subtask 1: Where does the input file(s) reside, in the local file system or in the HDFS file system? Why?

- The input files are in HDFS, because FileInputFormat.addInputPath(job, new Path(args[0])) tells Hadoop to read from the distributed filesystem for parallel processing.

Subtask 2: Where is the output file stored, in the local file system or in the HDFS file system? Why?

- The output directory (with part-r-00000) is in HDFS, because FileOutputFormat.setOutputPath(job, new Path(args[1])) writes the job result to the distributed filesystem.

Subtask 3: Where does your java program reside, in the local file system or in the HDFS file system? Why?

- The Java program's JAR (set with job.setJarByClass(WordCount.class);) resides on the local filesystem of the submit node, and Hadoop distributes it to worker nodes while keeping large data in HDFS.